

AICT_Lab15

Term_Project

Reg. No: 2026_(S)_CYS_12

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- 1) Clearly illustrate pseudo code for the complete design procedure of this task.**
- 2) Produce a python code to read a paragraph of at least 700 words from an input.**
- 3) Construct 11 separate functions.**

1. PSEUDOCODE:

```
FUNCTION count_words(text):
    count = 0
    split text into individual words
    for each word:
        add 1 to count
    return count

FUNCTION count_characters(text):
    count = 0
    for each character in text:
        add 1 to count
    return count

FUNCTION count_spaces(text):
    count = 0
    for each character in text:
        if character is a space:
            add 1 to count
    return count

FUNCTION count_commas(text):
    count = 0
    for each character in text:
        if character is a comma:
            add 1 to count
```

```

    return count
FUNCTION count_periods(text):
    count = 0
    for each character in text:
        if character is a period:
            add 1 to count
    return count
FUNCTION count_sentences(text):
    count = 0
    for each character in text:
        if character is a period, exclamation mark, or question
mark:
            add 1 to count
    return count
FUNCTION count_vowels(text):
    count = 0
    for each character in text:
        if character (lowercased) is a, e, i, o, or u:
            add 1 to count
    return count
FUNCTION count_digits(text):
    count = 0
    for each character in text:
        if character is a number (0-9):
            add 1 to count
    return count
FUNCTION count_uppercase(text):
    count = 0
    for each character in text:
        if character is an uppercase letter:
            add 1 to count
    return count
FUNCTION count_lowercase(text):
    count = 0
    for each character in text:
        if character is a lowercase letter:
            add 1 to count
    return count

```

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FUNCTION count_special(text):
    define a list of special characters (!@#$%^&* etc.)
    count = 0
    for each character in text:
        if character is in the special characters list:
            add 1 to count
    return count
print instructions to user
create empty list called lines
LOOP forever:
    read one line of input from user
    if line (uppercased) equals "END":
        join all lines into one text block
        count the words in text
        if word count is 700 or more:
            stop the loop
        else:
            tell user they need at least 700 words
    else:
        add the line to the lines list
print "Results:"
print Words           → call count_words(text)
print Characters      → call count_characters(text)
print Spaces          → call count_spaces(text)
print Commas          → call count_commas(text)
print Periods         → call count_periods(text)
print Sentences       → call count_sentences(text)
print Vowels          → call count_vowels(text)
print Digits          → call count_digits(text)
print Uppercase letters → call count_uppercase(text)
print Lowercase letters → call count_lowercase(text)
print Special characters → call count_special(text)

```

2. FUNCTIONS:

1. ***def count_words(text)***
2. ***def count_characters(text)***
3. ***def count_spaces(text)***
4. ***def count_commas(text)***
5. ***def count_periods(text)***
6. ***def count_sentences(text)***
7. ***def count_vowels(text)***
8. ***def count_digits(text)***
9. ***def count_uppercase(text)***
10. ***def count_lowercase(text)***
11. ***def count_special(text)***

3. CODE:

```
def count_words(text):  
    count = 0  
    for word in text.split():  
        count += 1  
    return count  
def count_characters(text):  
    count = 0  
    for char in text:  
        count += 1  
    return count  
def count_spaces(text):  
    count = 0  
    for char in text:  
        if char == " ":  
            count += 1  
    return count  
def count_commas(text):  
    count = 0
```

```
    for char in text:
        if char == ",":
            count += 1
    return count
def count_periods(text):
    count = 0
    for char in text:
        if char == ".":
            count += 1
    return count
def count_sentences(text):
    count = 0
    for char in text:
        if char in ".!?:":
            count += 1
    return count
def count_vowels(text):
    count = 0
    for char in text:
        if char.lower() in "aeiou":
            count += 1
    return count
def count_digits(text):
    count = 0
    for char in text:
        if char.isdigit():
            count += 1
    return count
def count_uppercase(text):
    count = 0
    for char in text:
        if char.isupper():
            count += 1
    return count
def count_lowercase(text):
    count = 0
    for char in text:
```

```

        if char.islower():
            count += 1
    return count
def count_special(text):
    special_characters = "!@#$%^&*()-_+=[]{}|;: '\",.<>?/\`~"
    count = 0
    for char in text:
        if char in special_characters:
            count += 1
    return count
print("Paste your paragraph below.")
print("When you are done, type END on a new line and press Enter.\n")
lines = []
while True:
    line = input()
    if line.strip().upper() == "END":
        text = "\n".join(lines)
        word_count = count_words(text)
        if word_count >= 700:
            break
        elif word_count < 700:
            print(f"\nNote: Atleast 700 words are required.\n")
    else:
        lines.append(line)
print("Results:")
print(f"Words           : {count_words(text)}")
print(f"Characters       : {count_characters(text)}")
print(f"Spaces           : {count_spaces(text)}")
print(f"Commas           : {count_commas(text)}")
print(f"Periods          : {count_periods(text)}")
print(f"Sentences        : {count_sentences(text)}")
print(f"Vowels           : {count_vowels(text)}")
print(f"Digits           : {count_digits(text)}")
print(f"Uppercase letters : {count_uppercase(text)}")
print(f"Lowercase letters : {count_lowercase(text)}")
print(f"Special characters: {count_special(text)}")

```